

# **K.A.L.A KCSE 2025 PREDICTION**

***(KENYA HIGH-ALLIANCE -LENANA-ALLIANCE GIRLS)***

## **BIOLOGY**

**231/3**

**PAPER 3 (PRACTICAL)**

**TIME: 1¾ HOURS**

**NAME.....**

**SCHOOL..... SIGN.....**

**INDEX NO..... ADM NO.....**

### ***Kenya Certificate of Secondary Education.***

#### **INSTRUCTIONS TO CANDIDATES**

- a) Write your name and Admission number in the spaces provided above.*
- b) Sign and write the date of examination in the spaces provided above*
- c) Answer **ALL** the questions in the spaces provided below each question.*
- d) Answer **all** the questions in the spaces provided.*
- e) All working **MUST** be clearly shown where necessary.*

#### **For Examiner's Use Only**

| <b>Question</b> | <b>Maximum Score</b> | <b>Candidate's Score</b> |
|-----------------|----------------------|--------------------------|
| <b>1</b>        | <b>14</b>            |                          |
| <b>2</b>        | <b>15</b>            |                          |
| <b>3</b>        | <b>11</b>            |                          |
| <b>TOTAL</b>    | <b>40</b>            |                          |

1. You are provided with a specimen **R** and the following reagents.

- Dilute HCL
  - Dilute Hydrogen Peroxide
  - Dilute Sodium Hydroxide
  - Three test tubes
- 10ml of measuring cylinder  
Scalpel  
3 labels  
Iodine solution

Using the above provisions, conduct the procedure below.

- Label the test tubes A, B, and C.
  - Using a scalpel obtain three cubes of side 1 cm from specimen R and reserve the rest.
  - Using a scalpel, chop the first and the second cubes into tiny pieces separately and place into the test tubes A and B respectively.
  - Place the remaining cube into test tube C as a whole.
  - Add  $2\text{cm}^3$  of the dilute HCL into the test tube A, then  $2\text{cm}^3$  of dilute Sodium Hydroxide into each of the test tubes B and C.
  - Add about  $5\text{cm}^3$  of Hydrogen Peroxide into each of the three test tubes A, B and C.
- a) Observe the amount of effervescence produced in each test tubes and complete the table below.

**(6 marks)**

| Test Tube | Contents | Amount of effervescence observed |
|-----------|----------|----------------------------------|
| A         |          |                                  |
| B         |          |                                  |

|   |  |  |
|---|--|--|
| C |  |  |
|---|--|--|

**B)** Explain the observation made in test tube **A** and **B**.

**(3 marks)**

**A:** .....

.....

**B:** .....

.....

**c)** What would be the expected result if **K** was first boiled before conducting the above procedures.

Result

**(2 marks)**

.....

Explanation:

.....

.....

**d)** Use the reagents provided to test for the food substance present in specimen **K** reserved from (a).

Observe and record in the table below.

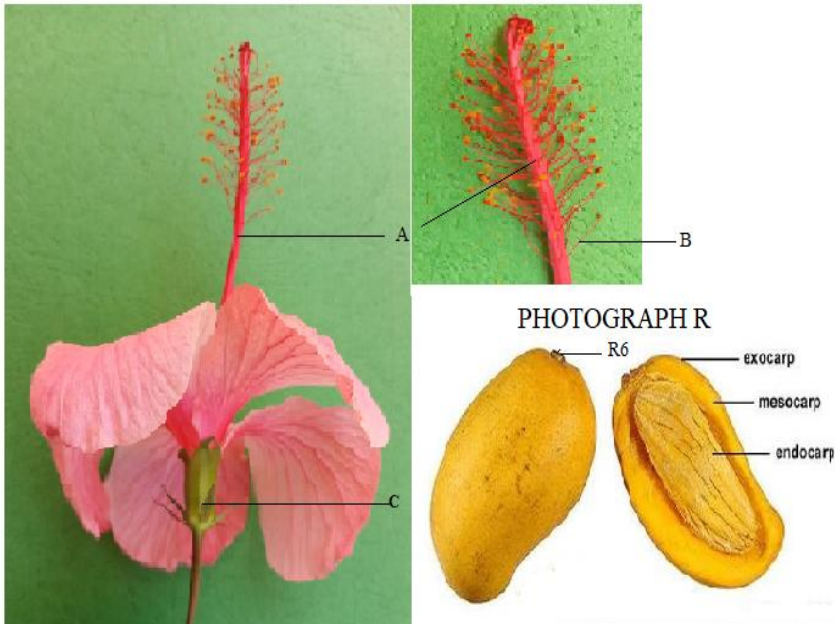
**(3 marks)**

| Procedure | Observation | Conclusion |
|-----------|-------------|------------|
|           |             |            |

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|

2. The photographs below are of certain plant organs. Use them to answer the questions that follow.

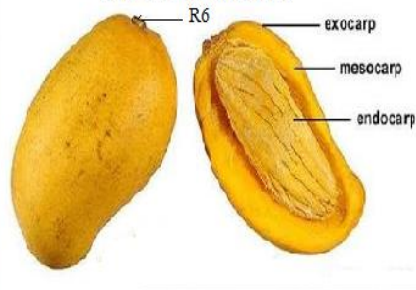
PHOTOGRAPH Q



PHOTOGRAPH T



PHOTOGRAPH R



a) (i) Identify the structures **A** and **B** in Photograph Q.

(2 marks)

**A:** .....

**B:** .....

ii) Give the role of structure **C**

(1 mark)

.....

iii) Using observable features only, state the agent of pollination for specimen **Q** giving a reason for your answer.

(2 marks)

Agent: .....

Reason: .....  
.....

b) Specimen in photograph **R** was cut longitudinally and displayed as shown, use it together with photograph **T** to answer the questions that follow

i) Give the functional difference between **R**<sub>6</sub> and **T**<sub>5</sub>.

(1 mark)

.....  
.....

ii) State the type of placentation in the two specimens in photographs **R** and **T** giving a reason in each case.

(4 marks)

**R:** .....

Reason: .....

.....

**T:** .....

Reason: .....

.....

iii) Giving **two** observable features, state the mode of dispersal of specimen **R**. (3 marks)

Mode: .....

Features: .....

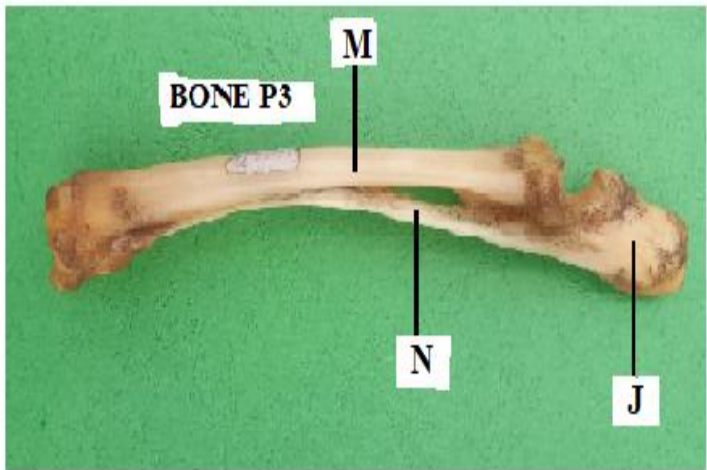
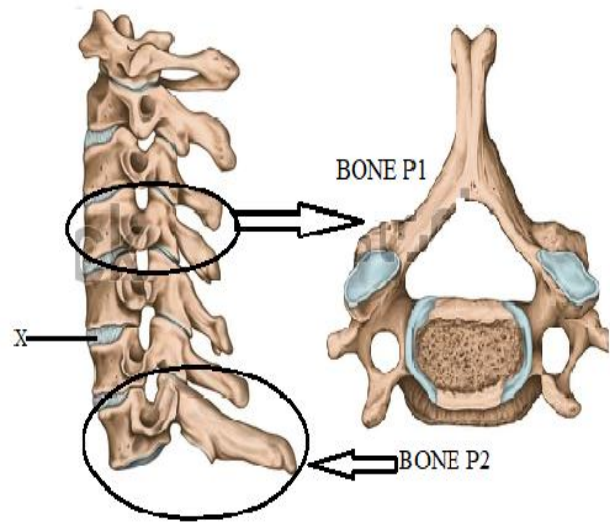
.....

iv) Identify **one** cause for seed dormancy displayed in **R** and state **one** way of breaking it. (2 marks)

.....

.....

3. The following are photographs of bones of the mammalian skeleton. Use them to answer the questions that follow.



a) Giving the main distinguishing feature in each case, identify the bones **P1** and **P2**. (4 marks)

**P1:** .....  
 Feature: .....  
 .....  
**P2:** .....  
 Feature: .....  
 .....

b) State **two** functions of the structure **X**. (2 marks)

.....  
 .....

c) Photograph **P3** is a composite set of two bones.

i) Give the identity of the parts labelled **M** and **N**. (2marks)

**M:** .....  
**N:** .....

ii) Name the bone that articulates with **P3** at its proximal end and state how the named is adapted to its function.

**(2 marks)**

Bone: .....

Adaptation:.....

.....

.....

iii) Other than muscle attachment, give the other function of the part labelled **J**.

**(1mark)**

.....

iv) Give **one** nutritional value of the human endoskeleton.

**(1 mark)**

.....

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